SCI-RTC

Raw-Timestream Conditioning and Temporal Response

Scientific Rationale

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Scope of this document. It explains the physical and statistical reasoning behind the shared SCI-RTC authority. It does not claim scientific approval, implementation conformity, representation fidelity, observational performance, validation, or production readiness.

Abstract

Raw-timestream conditioning is scientifically consequential even when it is described operationally as despiking, filtering, or downsampling. Each such operation can move signal between detectors or times, alter amplitude and phase, create correlations, change temporal support, and make a finite output depend on samples that are not independent astronomical measurements. SCI-RTC therefore treats conditioning as a realized detector-time operator, not as an array mutation followed by a nominal filter label.

The v0.1 contract is deliberately role-specific. The primary standardized Beammap input remains raw $\Delta f/f$; only a separately authorized CAL path may produce mJy/beam. When raw donor replacement is justified by the multiplicative convention $z_i = \texttt{flxscale}_i x_i$, its transfer is the donor factor divided by the target factor, with exact occurrence and domain validity. No legacy responsivity is required or revived. Phase-zero point selection is the sole downsampling operator. Any output influenced by ALIGN synthesis or RTC replacement is scientifically ineligible, while the numerical influence and its cause remain visible. A response is complete or explicitly unavailable.

Fixed and learned sampling share this authority. Learned mode may choose the maximum safe factor only after owner-resolved transfer, alias, sampling, realizability, and compatibility tests pass, using one immutable common observation plan. The output is an atomic bundle of signal, identity, support, response, influence, validity inputs, uncertainty status, provenance, diagnostics, and completion state. The appendices contain the normative core; the first ten Arabic-numbered pages are explanatory narrative.

> **Reading rule.** Sections 1–10 explain why the contract takes its form. Appendix A onward is the sole normative authority. An explanatory phrase cannot weaken an appendix requirement or resolve an open owner choice.

Contents

1 The scientific object is a conditioned measurement, not an array

The input to RTC is already an interpretation: an upstream authority has associated detector values with an ordered grid, declared which cells are original or synthesized, and supplied identities and support. RTC must not turn those bounded facts into stronger claims. In particular, a regular software grid is useful for filtering but does not by itself prove where a physical integration occurred or how accurately that event is tied to an absolute epoch. The distinction matters whenever a filter phase or a coordinate-dependent mask is interpreted astronomically.

Conditioning then adds another layer of interpretation. A despike detector selects cells from the data; a donor rule may move information across detectors; a finite filter couples neighboring times; an IIR carries state; a source mask changes the operator according to position; and a decimator keeps only selected points. The resulting number at one output slot is therefore not characterized by its value and unit alone. Its scientific meaning also includes where its information came from, which parts of that support were synthetic or replaced, how amplitude and phase changed, and whether the complete response can be reconstructed.

This motivates the boundary of the package. ALIGN continues to own grid and synthesis meaning. AST continues to own coordinate identity and validity. SCI-BEAM and SCI-CAL continue to own calibration-factor science. RTC owns only the ordered use of admitted inputs and the consequences of conditioning. Downstream packages retain their estimators and eligibility policies, but they may not strengthen an unavailable RTC fact. This ownership structure is not administrative decoration: it prevents a finite array from silently manufacturing timing, calibration, astrometry, covariance, or exposure.

The atomic output follows directly. Signal values, ordered identity, response, support, causes, validity inputs, uncertainty status, state, and completion are one scientific unit. If a required member cannot be written, the bundle has failed rather than partially succeeded. Optional detail may be absent only when that absence cannot alter values, eligibility, or failure.

> **Physical test.** Ask whether two equal output numbers with different donor histories, filter state, or synthesized support represent the same measurement. They do not. The bundle exists to preserve that difference.

2 Product role fixes signal meaning before processing

A digital filter preserves dimensions but need not preserve astronomical meaning. A low-pass output can still have the dimensions of its input while its point-source peak, phase, or effective beam-crossing response has changed. For that reason SCI-RTC attaches quantity, unit, sign, baseline/reference, and valid domain to the product role before any operator is applied. Unit alone is never the complete label.

The most important v0.1 separation is between raw and calibrated roles. The frozen Beammap path expects the detector observable in raw $\Delta f/f$. That raw stream is the primary standardized input to subsequent beam inference; substituting a calibrated timestream or map would change the scientific object even if the substitution appeared convenient. A different role may import an approved calibration operator and target mJy/beam, but the request must name that authority and the output must retain its lineage and response. Neither role inherits the other's factor, unit, or normalization.

This separation also clarifies what RTC does not own. The calibration factor may embody source, atmosphere, and beam information established elsewhere. RTC may apply the factor at a declared point in its sequence and must carry the resulting uncertainty and response consequences, but it cannot repair an invalid factor or infer a missing one. Conversely, a raw role may use calibration factors solely as an authorized cross-detector conversion without claiming that the output itself is calibrated. The factors then define a ratio between raw detector conventions, not a new output unit.

Several apparently harmless shortcuts are therefore scientifically distinct: using a sensitivity diagnostic as a signal multiplier; describing a filtered calibrated stream as photometrically unchanged because its unit string did not change; or reusing a factor from a row with the same number but a different detector occurrence. Each loses authority or identity. The contract instead requires explicit role binding and honest unavailability.

Role example	Authorized signal meaning	What cannot be inferred
Beammap	Raw detector $\Delta f/f$ with full RTC response	Calibrated flux, unchanged effective PSF, or absolute timing
CAL-authorized	Declared calibrated mJy/beam domain	Raw Beammap substitutability or preserved peak response from unit alone
Diagnostic-only	Exactly named diagnostic quantity and support	Scientific eligibility or policy effect unless separately selected

3 Calibration order and raw donor scaling

Cross-detector replacement is where signal domain and operator order meet. Suppose a raw value from donor detector q is used to represent target detector d. Under the admitted multiplicative convention, each detector's calibrated value equals its own `flxscale` times its raw value. The raw target replacement must therefore be multiplied by donor `flxscale` divided by target `flxscale`. Calibrating that replacement with the target factor then gives the same calibrated amplitude as calibrating the donor with the donor factor. The direction is fixed by this equality, not by a field name or historical convention.

The rule is intentionally narrow. Both factors must bind the exact detector occurrences, use the same calibration convention and domain, apply on the relevant support, and be valid; the target factor must be nonzero. A finite ratio between unrelated rows is not scientific authority. When any condition fails, the `flxscale` route is unavailable unless a different donor-to-target authority is separately admitted with direction, units, support, uncertainty, validity, and provenance.

This construction neither requires nor recreates legacy responsivity. That quantity has no canonical role in the frozen Beammap contract. Introducing it would add an unowned convention and could double-count or reverse a detector conversion. The clean statement is entirely in terms of the owner- modified multiplicative rule.

For a CAL-authorized role, calibration precedes cross-detector replacement. This makes the signal domain of the replacement explicit and avoids assuming that two detector-specific operations commute. A reversed order can be equivalent, but only as a full affine operator: factor direction, additive offsets, time dependence, response, uncertainty, identity, validity, and domain must all agree. Agreement for one constant or one fixture cannot establish that general equivalence.

Two limiting checks expose common mistakes. Equal compatible factors give a unit raw transfer. If the donor factor is two and the target factor is four, a donor raw value of three becomes a target raw value of one and a half; both calibrate to six. The reciprocal ratio gives twelve at the target and is immediately falsified. If the target factor is zero or invalid, no numerical answer is authorized.

Interpretive safeguard. A calibration factor may be used to relate two raw detector conventions without making the raw product calibrated. The role label, not the mere presence of a factor, controls output meaning.

4 Despiking is selection, mixing, and provenance

A despike operation has at least two scientific parts. First, a detector or rule decides that a target cell should not pass unchanged. Second, a replacement policy decides whether and how other cells contribute. Both parts may depend on the observed data, source masks, flags, detector topology, or local time support. The unconditional mapping is consequently not guaranteed to be linear, stationary, unbiased, or Gaussian, even when the replacement formula is affine after all choices are fixed.

The realized selection must therefore remain reconstructible. The output needs the target indicator, eligible candidate definition, donor score and tie rule, exact donor occurrences and time cells, weights, factor convention, fallback, and any additive baseline. Determinism at ties is scientifically important: otherwise identical data can acquire different detector mixing, response, and covariance solely from execution order. A fallback is also an operator. Leaving a spike unchanged, inserting zero, copying an invalid donor, or failing the product have different consequences and cannot share a single unlabeled state.

Replacement changes information, not just values. The target identity is retained because the output remains located at the target detector and time, but its support links every donor. Reusing a donor creates correlations between targets; filtering spreads that donor's influence into further outputs. None of these derived samples is a new independent exposure. A finite number can be useful for continuity while remaining unsuitable for a scientific estimator that assumes original independent measurements.

The v0.1 owner rule makes this consequence explicit and transitive. Any output influenced by RTC replacement is scientifically ineligible. The same is true for upstream ALIGN synthesis. The rule is not limited to the selected center of a downsampled output: a synthesized or replaced cell anywhere in the full filter or state support makes the output ineligible. Numerical influence is still retained in the response, and compact cause/support bookkeeping remains mandatory. Ineligibility must not erase the processing that actually occurred.

This separation allows downstream policy to remain honest. VAL can combine the causal facts with its own named rules; PTC or MAP can reject an affected sample without pretending the donor never influenced neighboring values. What they may not do is reinterpret finiteness or zero direct weight as independence.

Limiting case. When the target indicator is zero, replacement reduces to identity. When it is one, the donor link and influence are part of the measurement even if donor and target values happen to be equal.

5 Filtering requires state-complete response

For an ideal interior FIR with fixed coefficients, the familiar impulse and frequency responses are sufficient: coefficient sum gives DC gain, the complex response gives amplitude and phase, and its phase slope gives group delay where defined. This special case is valuable because impulses, constants, ramps, steps, and sinusoids provide strong deterministic falsification. They reveal reversed coefficients, normalization errors, wrong rate conventions, hidden delay, and stopband leakage without requiring an observational campaign.

Real conditioning often falls outside that ideal. An IIR or notch retains state, so its output depends on initialization and on where state is reset or carried. A forward/backward pass changes magnitude, effective order, and edges relative to a causal pass. A finite window needs padding, truncation, guarding, or rejection. A coordinate or data mask may split segments, renormalize weights, or hold state. Chained stages commute only in the translation-invariant convolution interior; at masks, boundaries, finite precision, and state transitions their order is observable.

The complete response must include those facts. A coefficient array by itself is not the response of a stateful, masked, detector-mixing, finite-window, multirate operator. The general object is the local detector-time Jacobian at fixed realized state, represented sparsely or as an exact factorization. It can include alignment weights, cross-detector donor links, calibration, each filter, state propagation, edges, and sampling. At a selection boundary the derivative may be discontinuous or undefined; this is an availability fact, not permission to publish a convenient scalar curve.

The complete-response-or-unavailable rule prevents partial truth from being misread as full truth. If a stored FIR kernel omits a donor mix or an IIR state, it may still be a useful component, but it is not the conditioned response. A downstream consumer may use the complete factorization or decline the affected claim. It may not assume omitted stages are identity.

The same reasoning applies to photometric labels. A filter can preserve mJy/beam dimension while attenuating a source crossing or moving its peak. Preserved dimensional unit is not preserved point-source response. The response and its uncertainty must travel with the signal.

Limiting case. As FIR coefficients approach a Kronecker delta, the interior filter approaches identity. As a causal pole approaches the unit circle, memory grows; stability cannot survive at or beyond the circle.

6 Masks, edges, and non-finite state are different causes

A source mask answers a processing question: should a selected operation use this coordinate-associated cell? It does not answer whether the acquisition exists, whether the coordinate is valid, whether the numerical value is finite, or whether a consumer may use the output. Conflating these questions turns scientific causes into accidental control flow.

Coordinate validity is especially important. A finite coordinate in the wrong frame, an out-of-domain TAN coordinate, an unavailable detector binding, and a valid coordinate outside a source region are distinct states. Only the last one is an outside-source result. Treating invalid coordinates as outside would admit precisely the samples for which mask membership cannot be known. The affected required operation must instead fail, retaining the cause. Pointing, OOF, and Beammap controls use their declared AltAz tangent plane; Science uses its declared J2000 TAN relation. RTC performs no quiet conversion between them.

Edges pose a parallel problem in time. A finite FIR needs a declared boundary extension or an output region with complete context. An IIR needs an initial state and a state/reset boundary; a finite guard approximates a longer tail only under a stated residual criterion. A short or empty scan is not an ordinary scan with fewer samples. These cells may be guarded, edge-invalid, conditionally available, or rejected according to selected policy, but they cannot be mislabeled as interior.

Non-finite values add state propagation. A NaN in an FIR affects a bounded footprint unless an earlier authorized operation replaces it. A NaN entering an IIR can poison the state-dependent tail until a declared recovery or reset establishes a new finite state. Silent zero conversion changes both response and statistical meaning. The contract therefore requires rejection, exact unavailable influence, or an explicitly selected prior recovery.

Flags remain descriptive causes. A flag may inform a mask or a downstream eligibility predicate only through a named policy. Likewise, a mask does not become a confidence, and zero weight does not erase exposure or influence. Keeping these axes separate allows a later consumer to reason about cause without reconstructing it from sentinels.

Falsification pattern. Test valid inside/outside coordinates beside invalid, unavailable, wrong-frame, and ambiguous cases; then inject NaN into samples, coefficients, coordinates, and state separately. A single generic "bad" path cannot preserve the required distinctions.

7 Phase-zero selection fixes output identity but not support

V0.1 authorizes one downsampling operator: select pre-sampling points whose input indices are integer multiples of the factor, beginning at phase zero. This deliberately excludes arithmetic means and other block aggregates. A point selector has a simple output index and representative assigned-grid time, which makes phase and identity auditable. It does not average away high-frequency content and is not an anti-alias filter.

The cardinality follows directly from the selected indices. An empty segment has no output; otherwise the sequence includes zero and every factor-th input index still smaller than the segment length. Odd and even factors require no special midpoint convention because the representative time is the selected point time. An edge policy may reject a selected candidate for incomplete context, but it cannot shift the point to a new phase or rename a block mean as phase-zero selection.

Output identity and scientific support must nevertheless remain separate. A selected value normally depends on the prefilter footprint around its center; earlier filters expand that footprint, an IIR extends it to a state boundary, and donor or alignment mappings can introduce cross-detector or additional time support. Flags and influence must aggregate over this transitive support. Center-only propagation would miss a replaced or synthesized neighbor that contributed to the selected value and would violate the ineligibility rule.

Multirate response makes the same point in frequency space. Decimation folds multiple input bands into each output frequency. The exact phase-zero identity contains every folded image after the realized prefilter. An alias-free claim therefore needs a stated input band, stopband metric and bound, and sufficient attenuation in every image. Unknown high-frequency content makes the response conditional; point selection cannot remove it.

The software grid's rate family remains conditional authority. Dividing its rate by the integer factor defines the output software cadence, not a new claim about physical detector integration timing. Output records must carry the parent grid and phase so a consumer cannot confuse those claims.

Limiting case. $M = 1$ selects every admitted point and leaves the software rate unchanged. The full RTC chain is identity only if all earlier stages, states, and affine terms are also identity.

8 Learned sampling is a bounded plan-selection problem

Fixed and learned sampling differ in how the plan is obtained, not in what an applied plan must mean. Both ultimately provide an exact filter, integer factor, phase-zero rule, rate, support semantics, compatibility state, and provenance before apply. Apply executes an immutable plan; it does not learn, retune, or silently fall back while producing science data.

The first learned scope asks for maximum safe reduction from a finite admitted candidate set. "Safe" is a conjunction, not a score that permits one failure to compensate for another. The astronomical transfer must stay within an owner-resolved tolerance over an owner-resolved signal band. Every phase-zero alias image must meet its bound. Sampling must remain adequate for the fastest valid in-scan crossing of the smallest admitted beam. The exact filter must be realizable under the selected family and precision. Every required downstream consumer must accept the common cadence and response. Only after all predicates pass is the largest factor selected.

The smallest beam and maximum valid in-scan speed make the plan conservative over the admitted observation. A percentile speed is useful diagnostic context, but it can miss a short fast crossing and therefore cannot be safety authority. Similarly, the beam-times-filter calculation must use the exact realized FIR response and exact phase-zero multirate response; a nominal cutoff label is insufficient. Source injection may later test a realization, but the analytical operator defines admission.

One common observation plan avoids cadence changes across arrays or scans in this first scope. Per-array or per-scan optimization, noise-aware choices, heterogeneous time grids, and continuous adaptation could be scientifically valuable, but they change downstream identity, covariance, and comparison semantics and require successor authority.

The state lifecycle protects reproducibility. The request, bootstrap facts, learned candidate evaluation, resolved plan, and applied state remain distinct. Restart restores the complete resolved plan and rejects an incompatible beam, speed support, scan set, coefficient precision, rate, phase, role, or parent. A learn-to-apply transfer is not evidence of convergence or observational performance.

Open authority. The selection equation is complete only after the scientific owner resolves the numerical tolerances, support estimators, candidate/filter family, fallback, and compatibility predicates listed in the owner ledger. Until then learned application is unavailable.

9 Covariance follows information flow, not storage shape

Once selections and states are fixed, the realized operator provides a clean conditional covariance statement: propagate the admitted full input covariance through the same detector-time map that produced the signal. This automatically captures correlations from reused donors, overlapping FIR support, IIR memory, alignment interpolation, and decimation footprints. A rectangular output array does not make its cells independent.

The statement is conditional because selectors are often learned from the same noisy data. A spike detector, donor choice, mask, fitted coefficient, or data-derived state can preferentially select particular noise realizations. Conditioning on the realized choice removes that randomness from the local calculation; it does not prove that selection bias or selection covariance is zero. Unconditional moments require the joint law of samples and selectors. If that law is absent, those terms are explicitly unavailable.

Calibration and response introduce another class of uncertainty. A common calibrator scale, atmosphere model, beam estimate, detector factors, donor ratios, filter coefficients, or uncertain cadence may correlate many samples or detectors. Such common terms do not average down as independent noise. Linearized nuisance propagation is appropriate only when the model is differentiable and the approximation is adequate; otherwise nonlinear propagation or a separately approved ensemble is needed. Cross terms are required when nuisance estimates share data with the timestream.

These distinctions explain why inverse variance, precision, response, and confidence cannot be synonyms. The reciprocal of one marginal conditional variance is not the diagonal of the full precision matrix. A scalar weight does not encode cross-detector covariance. A response curve does not say how uncertain that curve is. A confidence score does not establish exposure.

Unknown covariance is therefore unavailable, not diagonal or zero. A bounded consumer may accept a conditional weight while explicitly excluding selection and systematic terms, but it cannot use that weight for an absolute significance or total-error claim. The bundle records which statement is actually supported.

Limiting case. One donor reused twice creates shared random input in both targets. Even if both outputs are finite and stored in separate columns, their covariance is generally nonzero and neither is new independent exposure.

10 Atomic handoff preserves causal and claim boundaries

The conditioned TOD is useful only when a consumer can bind it to the exact observation, detector occurrences, output grid, scan and interval identities, role domain, parents, and processing stage. It also needs the complete response or an unavailable state; full support and transitive influence; typed acquisition, synthesis, replacement, mask, edge, filter, non-finite, and sampling causes; separated validity inputs; and the available statistical statement. These components form one atomic handoff because a mismatch among them can be more damaging than a missing product.

Compactness is compatible with exactness. Coefficients, state, interval exceptions, donor mappings, and generative support rules can reconstruct the operator without storing a dense response or provenance matrix for every sample. Compact representation ceases to be adequate when an exception cannot be reproduced, a cause is lost, or a partial kernel is mistaken for complete response. Required write failure propagates to completion rather than leaving an apparently valid TOD beside stale metadata.

Diagnostics follow the same discipline. Each diagnostic has a named estimator, unit, support, selection, validity, and availability. It is inert, advisory, or part of a selected policy. Optional detail is scientifically inert: toggling it cannot change values, flags, eligibility, response, cardinality, or failure. If a diagnostic affects a learned plan, it is no longer merely observational and must appear in the state and response.

Downstream ownership remains clear. PTC may clean and form weights only from an admitted bundle and may not treat influenced cells as independent. VAL may apply a named eligibility policy without collapsing causes. MAP and BEAM must bind the exact RTC parent and preserve response limitations. When PTC is disabled, a terminal RTC bundle may be the requested product and does not imply a PTC or map result.

Finally, the contract separates claims. Algebraic coherence can be reviewed from this document. Implementation conformity requires an independent trace. Representation fidelity requires evidence that stored artifacts reproduce the operator. Observational performance and science impact require their own data and criteria. Production readiness requires still different authority. No compilation or deterministic fixture collapses those layers.

Completion test. If a consumer can see the TOD but cannot tell whether a response-changing donor, edge, state, or sampling phase is present, the required RTC product is incomplete even when every number is finite.

A Shared normative authority and notation

Normative core stamp: `SCI-RTC-v0.1/r0.1`. The six files in `src/common/` form one authority. Text outside this shared core is explanatory or conformance guidance and cannot alter a definition, equation, assumption, requirement, prediction, unavailable state, or owner decision.

SCI-RTC begins with an admitted aligned primary detector stream and ends with an atomic, role-specific conditioning bundle. ALIGN, AST, SCI-BEAM, and SCI-CAL retain authority for the upstream meanings they produce. PTC, VAL, MAP, BEAM, NOI, MAP-003, FLT, SRC/MODE, and FRUIT retain authority for their downstream science. This contract owns the realized conditioning order and its scientific consequences; it does not infer current software behavior.

The words **shall** and **shall not** identify requirements. **Required unavailable** means that the affected product or claim shall not be reported as complete. **Optional unavailable** is a typed, scientifically inert state. Missing, disabled, automatic, rejected, invalid, failed, unavailable, unsupported, and omitted remain distinct states.

A.1 Indices, sets, and role labels

Symbol	Normative meaning
o	Immutable observation identity.
$d, q \in \mathcal{D}_o$	Exact admitted detector occurrences; d is a target and q may be a donor. A detector index or row number alone is not this identity.
r	Native acquisition row; it is not an aligned or conditioned slot.
$j \in \{0, \dots, N-1\}$	Ordered aligned input slot on the admitted ALIGN grid.
$n \in \{0, \dots, N_y - 1\}$	Ordered phase-zero RTC output slot.
$\mathcal{R}$	Product role, including Pointing, OOF, Science, Beammap, diagnostic-only, or another separately approved role.
$\delta_{\mathcal{R}}$	Role-domain selector: 0 for an admitted raw role and 1 for a separately CAL-authorized calibrated role.
x^A_{dj}	Admitted aligned primary `xs` value with declared physical observable, sign, baseline/reference, unit, support, and validity.
z_{dj}	Calibrated value when the role admits an imported CAL operator.
u_{dj}	Role-domain value immediately before the ordered RTC filter chain.
y_{dn}	Conditioned role-domain output at the selected phase-zero slot.
ϕ_d	`flxscale` for one exact detector occurrence under the declared multiplicative convention and calibration domain.
A_α	Imported, realized ALIGN mapping with state α; its scientific meaning is not owned by RTC.
C_κ	Imported CAL operator with state κ; it is the identity for a raw role.
B_ω	Realized despike/replacement operator conditioned on selection state ω.
F_{k,θ_k}	Realized ordered FIR, IIR, notch, or admitted mask-controlled stage k.
$D_{M,0}$	Integer phase-zero point selector of factor $M \geq 1$.
Ω	Complete realized conditioning state, including all upstream bindings and every selected stage, exception, state, edge rule, and output interval.
$K_{dn,qj}$	Local detector-time response of one output cell to one acquired input cell.
$\mathcal{S}_{dn}$	Exact transitive input support of output (d, n), including alignment, donors, filters, state, and selection.
$\mathcal{I}_{dn}$	Cause-preserving transitive influence set for output (d, n).

A.2 Identity and interval conventions

Observation, Tune, array, network/interface, detector occurrence, selected APT row/binding, native row, aligned slot, scan, science interval, context interval, conditioned output slot, stage, and product identities are non-interchangeable. Array identity is one of `a1100`, `a1400`, and `a2000`; array ID, array index, network ID, detector index, and container position do not substitute for one another. Cross-artifact identity requires an explicit occurrence- and product-scoped binding. Equal row numbers do not establish identity.

Primary detector timestreams are samples on rows and detectors on columns unless a product explicitly declares another ordered shape. Every interval is half-open. Science windows, physical scans, processing chunks, filter context, output selections, and learned-plan scan sets retain separate identities. Native, aligned, and output indices never silently renumber one another.

Time is seconds on the admitted ALIGN-assigned grid. The approximately 8-ms rate relationships and the $0.5\times$, $2\times$, and $4\times$ family are conditional software-grid facts, not proof of a detector integration event, absolute epoch, or astrometric correction. Coordinate-dependent Pointing, OOF, and Beammap controls use a declared AltAz tangent plane; Science controls use a declared equatorial J2000 TAN relation. RTC performs no implicit frame conversion.

A.3 Logical and availability notation

$\mathbb{I}[\cdot]$ denotes an indicator, $\mathbb{E}$ expectation, Cov covariance, and $\mathrm{supp}(\cdot)$ operator support. The logical symbols $\wedge$, $\vee$, and $\neg$ have their usual meanings. A quantity decorated available is present with its required identity, domain, and validity; a quantity decorated unavailable is explicitly unavailable. Neither zero nor NaN is an availability encoding.

End of notation; shared definitions follow without a change of authority.

B Normative definitions

SCI-RTC-DEF-001 – Admitted aligned stream.
An immutable observation-scoped primary `xs` array whose physical observable, sign, zero/reference, baseline, unit, ordered detector binding, assigned sample grid, cadence, phase, support, origin/synthesis state, validity, scan partition, and mapping lineage are supplied by the responsible upstream authority. Admission does not upgrade unavailable physical event timing.

SCI-RTC-DEF-002 – Product-role signal domain.
The tuple $(\mathcal{R}, Q_{\mathcal{R}}, U_{\mathcal{R}}, s_{\mathcal{R}}, b_{\mathcal{R}}, \mathcal{V}_{\mathcal{R}})$ naming role, physical quantity, unit, sign, reference/baseline, and valid domain. Frozen Beammap uses raw $\Delta f/f$. A separately CAL-authorized role may use calibrated mJy/beam. The domains never substitute silently.

SCI-RTC-DEF-003 – Exact detector occurrence.
An observation-local layered binding containing observation, Tune, network/interface, network-local tone or admitted row slot, selected APT identity/digest, and row association. A transient column number is only a locator.

SCI-RTC-DEF-004 – Imported CAL operator.
An immutable SCI-CAL-selected transformation carrying its detector binding, source and target atmosphere lineage, target unit, factor convention, validity, uncertainty, availability, and provenance. RTC may apply it in an authorized role and owns that order; RTC does not derive, repair, or promote the factor.

SCI-RTC-DEF-005 – Compatible `flxscale` pair.
Two finite factors ϕ_q and ϕ_d that are valid for the exact donor and target detector occurrences under the same multiplicative calibration convention, quantity domain, reference plane, and applicable support, with $\phi_d \neq 0$. Compatibility is not established by matching row numbers or finite values alone.

SCI-RTC-DEF-006 – Selected conditioning policy.
A versioned, admitted specification of despike detection, donor eligibility, donor scoring and ties, replacement and fallback, source protection, filter stages and order, exact coefficients and normalization, numeric precision,

state/reset, boundaries/edges, non-finite handling, sampling mode, phase-zero factor, flags, influence aggregation, diagnostics, and failure behavior. It is selected authority, not an implementation default.

SCI-RTC-DEF-007 – One-way state record.
The ordered record $(\Theta^{\mathrm{req}}, \Theta^{\mathrm{eff}}, \Theta^{\mathrm{obs}}, \Theta^{\mathrm{learned}}, \Theta^{\mathrm{real}})$. Requested state preserves the accepted request; effective state is the admitted typed plan; observation-resolved state binds exact observation facts; learned/resolved state is present only in learned mode; and realized state records what ran. A later state never rewrites an earlier state.

SCI-RTC-DEF-008 – Realized conditioning operator.
The chronological composition of every admitted alignment, role-domain, replacement, mask, FIR/IIR/notch, boundary, state, non-finite, and phase-zero sampling action, including all affine state terms and exception intervals. Annotations that affect values, support, influence, or availability are part of the operator.

SCI-RTC-DEF-009 – Phase-zero sampling.
Point selection of the pre-sampling stream at input indices $j = Mn$ for one integer factor $M \geq 1$. It is not an arithmetic mean, block aggregate, or implicit anti-alias filter.

SCI-RTC-DEF-010 – Complete realized response.
An exact local or factorized representation that includes every enabled response-changing stage, detector mixing, mask, state, edge rule, exception, and sampling phase on its declared support. If such a representation is not truthfully available, response state is unavailable; a partial kernel is not complete response.

SCI-RTC-DEF-011 – Direct source cell.
An aligned cell that enters an RTC value before transitive propagation. It may be original, ALIGN-synthesized, or RTC-replaced. Numerical participation does not confer independent scientific exposure.

SCI-RTC-DEF-012 – Transitive influence.
The cause-preserving closure from every original, synthesized, replaced, masked, edge, state, and non-finite source cell through all nonzero realized alignment, donor, filter, state, and sampling dependencies to an output. Influence is broader than the selected point and may cross detector columns.

SCI-RTC-DEF-013 – Scientifically ineligible RTC output.
An output having any transitive influence from an ALIGN-synthesized or RTC-replaced cell. It remains ineligible even when finite and even when the influenced cell is not the phase-zero selected center. Causes and numerical influence remain reportable.

SCI-RTC-DEF-014 – Coordinate-dependent control.
A source mask or response operation that consumes an AST coordinate identity, frame, topology, detector binding, validity, uncertainty where required, and support. Invalid or unavailable coordinates are not outside-source values.

SCI-RTC-DEF-015 – Fixed sampling mode.
A request whose admitted factor, realizable prefilter, rate, phase-zero rule, and downstream compatibility are resolved without learning them from the observation.

SCI-RTC-DEF-016 – Learned sampling mode.
An optional request that evaluates an admitted finite candidate set and resolves one immutable observation-wide cadence/filter plan before apply. Its admission uses the smallest admitted beam, maximum valid in-scan speed, analytical beam-times-realized-filter response, exact phase-zero alias accounting, sampling sufficiency, filter realizability, and downstream compatibility. Noise-aware, per-array, per-scan, heterogeneous-cadence, and continuous adaptation are outside v0.1.

SCI-RTC-DEF-017 – Conditional statistical covariance.
The covariance obtained from an admitted input covariance through a fixed realized operator while holding selection and nuisance state fixed. It is not total uncertainty and does not imply a diagonal, white, stationary, or independent noise model.

SCI-RTC-DEF-018 – Atomic RTC bundle.
One role-specific completion unit containing conditioned TOD; ordered detector/sample/scan/support and parent identity; complete response or typed unavailability; cause-preserving flags and influence; separated validity and eligibility inputs; conditional uncertainty/covariance status; one-way state and provenance; required diagnostics; and completion/failure state. TOD alone is not this bundle.

SCI-RTC-DEF-019 – Scientifically named diagnostic.
A quantity with an explicit estimator, unit, support, selection/mask, validity, uncertainty/availability, and policy role. It is either inert, advisory, or an input to an explicitly selected policy; those roles are not interchangeable.

SCI-RTC-DEF-020 – Claim layer.
One of algebraic contract correctness, implementation conformity, representation fidelity, observational perfor-

mance, science-impact qualification, or production readiness. Evidence at one layer does not establish another.

C Normative equations and identities

The equations in this section are the sole displayed mathematical authority of both views. They describe an implementation-independent, state-conditioned operator. They neither select numerical policy nor assert implementation conformity.

C.1 Role domain and donor transfer

For an exact detector occurrence i under the admitted multiplicative raw-to-calibrated convention,

$$z_i = \phi_i x_i, \qquad \phi_i = \texttt{flxscale}_i. \tag{SCI-RTC-EQ-001}$$

Equation (SCI-RTC-EQ-001) does not assign a role to legacy `responsivity`. When q is a raw donor and d is a raw target, the unique multiplicative transfer that represents donor-calibrated amplitude in the target raw convention is

$$x^R_{d\leftarrow q} = a_{d\leftarrow q} x_q, \qquad a_{d\leftarrow q} = \frac{\phi_q}{\phi_d}, \qquad \phi_d x^R_{d\leftarrow q} = \phi_q x_q. \tag{SCI-RTC-EQ-002}$$

This equation is defined only for a compatible `flxscale` pair as defined above. Otherwise this transfer is unavailable. An additive calibration convention, a different unit/reference domain, or an alternative donor authority requires its own complete affine equivalence and is not silently folded into equation (SCI-RTC-EQ-002).

Let the role-domain operator be

$$C_\mathcal{R} = \begin{cases} I, & \delta_\mathcal{R} = 0 \quad \text{(admitted raw role)}, \\ C_\kappa, & \delta_\mathcal{R} = 1 \quad \text{(CAL-authorized role)}. \end{cases} \tag{SCI-RTC-EQ-003}$$

For a CAL-authorized role, calibration precedes cross-detector replacement. For a raw role, $C_\mathcal{R} = I$ and an admitted raw donor transfer may use equation (SCI-RTC-EQ-002). Reversing calibration and replacement is equivalent only if, for the complete admitted affine operators on the exact support,

$$B^{(z)}_\omega C_\kappa = C_\kappa B^{(x)}_\omega, \quad \mathbf{b}^{(z)}_\omega = C_\kappa \mathbf{b}^{(x)}_\omega + \mathbf{c}_\kappa - B^{(z)}_\omega \mathbf{c}_\kappa, \tag{SCI-RTC-EQ-004}$$

with unit, factor direction, additive offset, response, uncertainty, identity, validity, and domain equivalence all proved. Equality of numerical values in one fixture is insufficient.

C.2 Realized conditioning operator

Conditioned on complete realized state Ω, the ordered role-specific operator is

$$\boxed{\mathbf{y}_\mathcal{R} = D_{M,0} F_{K,\theta_K} \cdots F_{1,\theta_1} B_{\omega,\mathcal{R}} \left[C_\mathcal{R} (A_\alpha \mathbf{x}^{\text{acq}} + \mathbf{a}_\alpha) + \mathbf{c}_\mathcal{R} \right] + \mathbf{b}_\Omega\ .} \tag{SCI-RTC-EQ-005}$$

Chronological order is read from right to left. The affine terms include only declared calibration, replacement, filter-state, boundary, or other authorized contributions. Masks, coefficients, finite handling, context, exceptions, and state are components of Ω, not annotations outside the operator. Conditional on them,

$$\mathbf{y}_\mathcal{R} = L_\Omega \mathbf{x}^{\text{acq}} + \mathbf{b}'_\Omega, \qquad L_\Omega = D_{M,0} F_K \cdots F_1 B_{\omega,\mathcal{R}} C_\mathcal{R} A_\alpha. \tag{SCI-RTC-EQ-006}$$

This is a conditional affine identity, not an unconditional linear, stationary, unbiased, or Gaussian claim.

A realized replacement at target cell (d, j) may be written

$$u^R_{dj} = (1 - p_{dj}) u_{dj} + p_{dj} \left[\sum_{(q,\ell)\in\mathcal{Q}_{dj}} w_{dj,q\ell}\, a_{d\leftarrow q,j\ell}\, u_{q\ell} + \beta_{dj} \right], \quad \sum_{q,\ell} w_{dj,q\ell} = 1 \text{ when normalization is claimed.} \tag{SCI-RTC-EQ-007}$$

The scale a is domain-qualified: equation (SCI-RTC-EQ-002) is the authorized raw `flxscale` route, while a calibrated-domain transfer or any separately authorized route carries its own identity and validity.

For a time-invariant FIR stage,

$$v_{d,j} = \sum_{\ell \in \mathcal{L}_k} h^{(k)}_{d,\ell} u_{d,j-\ell}, \qquad H^{(k)}_d(e^{i\omega}) = \sum_{\ell \in \mathcal{L}_k} h^{(k)}_{d,\ell} e^{-i\omega\ell}, \qquad H^{(k)}_d(1) = \sum_{\ell} h^{(k)}_{d,\ell}. \tag{SCI-RTC-EQ-008}$$

For a causal IIR or notch stage, a state-complete form is

$$\mathbf{s}_{j+1} = A_k \mathbf{s}_j + B_k u_j, \qquad v_j = C_k \mathbf{s}_j + D_k u_j, \qquad \mathbf{s}_{j_0} = \mathbf{s}_{0,k}. \tag{SCI-RTC-EQ-009}$$

An exactly equivalent recurrence is admissible only with its coefficient convention, precision, state, and boundary identity. A causal-stability claim requires every realized pole strictly inside the unit circle.

One possible selected mask operator is normalized masked FIR convolution,

$$v_j = \frac{\sum_\ell h_\ell m_{j-\ell} u_{j-\ell}}{\sum_\ell h_\ell m_{j-\ell}}, \qquad \left| \sum_\ell h_\ell m_{j-\ell} \right| > \epsilon_m, \tag{SCI-RTC-EQ-010}$$

but equation (SCI-RTC-EQ-010) is not a selected default. Segmenting, zero filling, restoring a mask after filtering, holding state, and renormalizing are distinct operators.

C.3 Phase-zero sampling and response

For a coherent pre-sampling segment of length N, v0.1 selects only

$$y_{d,n} = u_{d,Mn}, \qquad 0 \le Mn < N, \qquad f_{\text{out}} = \frac{f_{\text{in}}}{M}, \qquad N_y = \begin{cases} 0, & N = 0, \\ 1 + \left\lfloor \dfrac{N-1}{M} \right\rfloor, & N > 0. \end{cases} \tag{SCI-RTC-EQ-011}$$

The representative assigned-grid time is the selected point time $t^{\text{out}}_n = t_{Mn}$; the scientific support is nevertheless the complete transitive support $\mathcal{S}_{dn}$ of all preceding stages. A selected edge/output policy may reject candidate points, but shall not relabel a different phase or silently change equation (SCI-RTC-EQ-011).

For output (d, n) and acquired input (q, j), the exact local response is

$$K_{dn,qj} = \frac{\partial y_{dn}}{\partial x^{\text{acq}}_{qj}} \quad \text{at fixed realized state } \Omega. \tag{SCI-RTC-EQ-012}$$

At a selection boundary the derivative may be discontinuous or undefined; that state shall be represented rather than replaced by an LTI claim.

In the fixed, detector-separable, translation-invariant, single-rate interior case only,

$$H^{\text{comb}}_d(e^{i\omega}) = \prod_{k=1}^{K} H^{(k)}_d(e^{i\omega}), \qquad \tau_g(\omega) = -\Delta t \,\frac{d}{d\omega}\,\text{unwrap}\big[\arg H^{\text{comb}}_d(e^{i\omega})\big]\,. \tag{SCI-RTC-EQ-013}$$

Amplitude is $|H|$, power response is $|H|^2$, and phase is $\arg H$ under the recorded Fourier convention. Group delay is unavailable at response zeros unless a bounded convention is supplied.

For an LTI prefilter followed by phase-zero selection, the exact multirate identity is

$$Y(e^{i\Omega_o}) = \frac{1}{M} \sum_{r=0}^{M-1} H\Big(e^{i(\Omega_o + 2\pi r)/M}\Big)\, X\Big(e^{i(\Omega_o + 2\pi r)/M}\Big)\,, \tag{SCI-RTC-EQ-014}$$

under the declared Fourier/sign convention. Every folded image contributes; point selection itself is not anti-alias filtering.

For an FIR output, direct support starts with

$$\mathcal{S}^{(1)}_{dn} = \{Mn - \ell : \ell \in \mathcal{L}\}, \qquad \mathcal{S}_{dn} = \text{closure}_{A,B,F,\,\text{state}}\Big(\mathcal{S}^{(1)}_{dn}\Big)\,. \tag{SCI-RTC-EQ-015}$$

IIR support extends to the declared state/reset or context boundary and may be mathematically infinite. Any finite guard is an approximation with a stated residual-tail criterion.

C.4 Moments, influence, and learned-plan selection

For admitted input mean $\boldsymbol{\mu}_x$ and full covariance Σ_x, conditional on fixed Ω independent of the modeled random noise,

$$\mathbb{E}[\mathbf{y} \mid \Omega] = L_\Omega \boldsymbol{\mu}_x + \mathbf{b}'_\Omega, \qquad \text{(SCI-RTC-EQ-016a)}$$

$$\Sigma^{\text{stat}}_{y|\Omega} = L_\Omega \Sigma_x L_\Omega^{\mathsf{T}}. \qquad \text{(SCI-RTC-EQ-016b)}$$

If selectors depend on the random data, unconditional covariance obeys

$$\mathrm{Cov}(\mathbf{y}) = \mathbb{E}_\Omega[\mathrm{Cov}(\mathbf{y} \mid \Omega)] + \mathrm{Cov}_\Omega(\mathbb{E}[\mathbf{y} \mid \Omega]). \qquad \text{(SCI-RTC-EQ-017)}$$

The second term is not zero by default. With named nuisance parameters $\boldsymbol{\eta}$ and an adequate local linearization,

$$\Sigma^{\text{sys}}_y \approx J_\eta \Sigma_\eta J_\eta^{\mathsf{T}}, \qquad J_\eta = \left. \frac{\partial \mathbb{E}[\mathbf{y} \mid \Omega, \boldsymbol{\eta}]}{\partial \boldsymbol{\eta}} \right|_{\boldsymbol{\eta}_0}, \qquad \text{(SCI-RTC-EQ-018)}$$

and a total-covariance claim requires every applicable component and cross term:

$$\Sigma^{\text{total}}_y = \Sigma^{\text{stat}}_y + \Sigma^{\text{sys}}_y + \Sigma^{\text{selection}}_y + \Sigma^{\text{model}}_y + \Sigma^{\text{cross}}_y. \qquad \text{(SCI-RTC-EQ-019)}$$

Unknown components are unavailable, not zero.

Define cause-preserving influence and the v0.1 ineligibility rule by

$$\mathcal{I}_{dn} = \text{closure}_{K,\,\text{generative links}}\{\text{typed source causes on } \mathcal{S}_{dn}\}, \qquad \text{(SCI-RTC-EQ-020a)}$$

$$E^{\text{RTC}}_{dn} = V^{\text{det}}_d \wedge V^{\text{num}}_{dn} \wedge V^{\text{support}}_{dn} \wedge V^{\text{response}}_{dn} \wedge \neg\mathbb{I}[\text{ALIGN\_synth} \in \mathcal{I}_{dn}] \wedge \neg\mathbb{I}[\text{RTC\_replace} \in \mathcal{I}_{dn}]. \qquad \text{(SCI-RTC-EQ-020b)}$$

Consumer-specific coordinate or policy predicates may further restrict eligibility; they may not remove either transitive-influence exclusion.

For learned mode, let $\mathcal{M}$ be the admitted finite integer candidate set and $\mathcal{F}(M)$ the conjunction of all owner-resolved astronomical transfer, exact phase-zero alias, sampling, realized-filter realizability, and downstream-compatibility constraints, evaluated with the smallest admitted beam and maximum valid in-scan speed. Then

$$\mathcal{M}_{\text{safe}} = \{M \in \mathcal{M} : \mathcal{F}(M) = \text{true}\}, \qquad M_\star = \max \mathcal{M}_{\text{safe}}. \qquad \text{(SCI-RTC-EQ-021)}$$

Equation (SCI-RTC-EQ-021) is defined only after every numerical tolerance, support rule, fallback, and compatibility predicate named in the owner ledger is resolved. Percentile speed is diagnostic, not safety authority.

Finally, the atomic product is

$$\mathcal{O}_{\text{RTC},\mathcal{R}} = (\mathbf{y}, \mathcal{J}, \mathcal{K}, \mathcal{S}, \mathcal{I}, \mathcal{F}, \mathcal{V}, \mathcal{U}, \mathcal{P}, \mathcal{D}, \mathcal{C}), \qquad \text{(SCI-RTC-EQ-022)}$$

where $\mathcal{J}$ is ordered identity, $\mathcal{K}$ complete response status, $\mathcal{S}$ support, $\mathcal{I}$ influence, $\mathcal{F}$ typed flags/causes, $\mathcal{V}$ separated validity and eligibility inputs, $\mathcal{U}$ uncertainty/covariance availability, $\mathcal{P}$ one-way state/provenance, $\mathcal{D}$ diagnostics, and $\mathcal{C}$ atomic completion or required-output failure state.

D Bounded assumptions and non-goals

SCI-RTC-ASM-001 – Capability.
V0.1 treats Stokes-I primary `xs`. Enabled polarimetry and measured R-channel execution are excluded until separately approved.

SCI-RTC-ASM-002 – Aligned-grid boundary.
The input grid, cadence, phase, mapping, gaps, synthesis/origin state, support, and scan partition are admitted ALIGN facts. Physical event timing, absolute timing accuracy, and a producer epoch may remain unavailable.

SCI-RTC-ASM-003 – Coordinate boundary.
Coordinate-dependent operations assume an admitted AST identity, frame, topology, detector binding, support, validity, and uncertainty where required. The absence of such authority does not license inference or clamping.

SCI-RTC-ASM-004 – Calibration boundary.
Calibration-factor production, atmosphere lineage, beam meaning, uncertainty, and promotion belong to SCI-BEAM/SCI-CAL. RTC conditions on an admitted operator only for a role that authorizes it.
SCI-RTC-ASM-005 – Selected-policy boundary.
Despike, donor, mask, FIR, IIR, notch, state, edge, non-finite, recovery, sampling, diagnostic, and failure choices are selected versioned policies. No current default or numerical threshold is promoted to a universal scientific constant by this contract.
SCI-RTC-ASM-006 – Conditional linearity.
Linear response and covariance identities condition on fixed realized selection and state. Data-derived selectors make the unconditional operation generally nonlinear and nonstationary.
SCI-RTC-ASM-007 – Statistical input.
Moments or covariance are claimed only from a supplied statistical model with declared correlation and limitations. Absence does not imply diagonal, white, stationary, independent, or zero covariance.
SCI-RTC-ASM-008 – Learned-mode scope.
The first learned applied scope produces one immutable common observation plan. Per-array, per-scan, heterogeneous-cadence, noise-aware, and continuously adaptive execution are successor scopes.
SCI-RTC-ASM-009 – Analytical authority.
Exact analytical beam-times-filter transfer and phase-zero alias accounting define learned-plan admission. Deterministic vectors may falsify a realization; source injection is not required to define the operator.
SCI-RTC-ASM-010 – Compact representation.
Dense observation-sized response, covariance, or per-sample provenance is not required when a factorized, sparse, interval-based, or generative representation exactly reconstructs the operator, support, causes, and exceptions.
SCI-RTC-ASM-011 – Downstream authority.
RTC supplies causal inputs but does not own PTC cleaning/weights, VAL eligibility policy, MAP estimation, BEAM inference, empirical NOI products, MAP-003 response tracing, map filtering, source fitting, or fruit-loop recurrence.
SCI-RTC-ASM-012 – Claim separation.
The contract is an algebraic and semantic authority only. It contains no claim that a present implementation conforms, that a representation is faithful, that observational performance is adequate, or that a mode is ready for production.

D.1 Explicit non-goals

This revision does not derive ALIGN timing, AST coordinates, SCI-BEAM or SCI-CAL calibration, PTC, VAL, MAP, or FLT estimators, source fitting, or recurrence; authorize absolute-timing or astrometric corrections; redesign mature despike/FIR/IIR/notch numerics; choose universal numerical thresholds; activate polarimetry or measured R; prescribe source classes, files, storage schemas, or product names; authorize per-array/per-scan cadence; inspect implementation; perform validation or reductions; or assert production status.

E Sequential normative requirements

The following identifiers are stable within v0.1/r0.1. Conformance is requirement-by-requirement; satisfying a neighboring requirement does not waive another one.

Area	ID	Normative statement
Boundary	`SCI-RTC-REQ-001`	RTC shall admit only an aligned primary `xs` stream whose signal meaning, ordered identity, assigned grid, support, origin/synthesis state, and validity are declared.
Boundary	`SCI-RTC-REQ-002`	Every requested product role shall declare physical quantity, unit, sign, reference/baseline, valid domain, and required consumers; no channel or role shall inherit another role's meaning.
Boundary	`SCI-RTC-REQ-003`	The primary standardized Beammap detector signal shall remain raw $\Delta f/f$ and shall not be silently replaced by a calibrated timestream or map.
Boundary	`SCI-RTC-REQ-004`	A calibrated mJy/beam domain shall be produced only for a separately CAL-authorized role with admitted factor identity, convention, atmosphere lineage, detector binding, validity, uncertainty availability, and provenance.
Boundary	`SCI-RTC-REQ-005`	RTC shall own the exact ordered application and response consequence of an imported CAL operator but shall not derive, repair, promote, or reinterpret its factor science.
Identity	`SCI-RTC-REQ-006`	Observation, Tune, array, network/interface, detector occurrence, APT binding, native row, aligned slot, scan, science interval, context interval, output slot, stage, and product identity shall remain distinct and immutable.
Identity	`SCI-RTC-REQ-007`	The ordered detector and sample shapes, index bases, native-to-aligned mapping, aligned-to-output map, scan membership, and memory order shall be explicit; indices shall not serve as cross-artifact detector identity.
Identity	`SCI-RTC-REQ-008`	RTC shall preserve the admitted ALIGN cadence, phase, rate, gaps, support, and synthesis/origin state without claiming physical integration-event timing or absolute epoch authority.
Identity	`SCI-RTC-REQ-009`	A coordinate-dependent operation shall consume the exact AST identity, frame, topology, detector binding, support, and validity required by its product role, with no implicit frame conversion.
State	`SCI-RTC-REQ-010`	Requested, effective, observation-resolved, learned/resolved when applicable, and realized state shall be recorded separately in one-way order; later state shall not overwrite the accepted request.
State	`SCI-RTC-REQ-011`	Outer, inner, full, mini, diagnostic, simulated, processed, and any other produced stage roles shall have distinct immutable identities and explicit parent/processing links without requiring duplicate computation.
Policy	`SCI-RTC-REQ-012`	Every despike, donor, replacement, mask, filter, state, edge, non-finite, recovery, sampling, flag, diagnostic, and fallback action shall be governed by an admitted versioned selected policy rather than an undocumented default.
Order	`SCI-RTC-REQ-013`	For a CAL-authorized role, calibration shall precede cross-detector replacement unless complete affine equivalence in equation (SCI-RTC-EQ-004) is proved for the exact unit, factor, offset, response, uncertainty, identity, validity, and domain.

Area	ID	Normative statement
Donor	`SCI-RTC-REQ-014`	A raw donor q to raw target d using the multiplicative convention $z_i = \phi_i x_i$ shall use ϕ_q/ϕ_d only when both `flxscale` factors are valid for the exact occurrences under the same compatible convention/domain and $\phi_d \neq 0$.
Donor	`SCI-RTC-REQ-015`	The raw donor route in REQ-014 shall use `flxscale` and shall neither require nor restore a scientific role for legacy `responsivity`.
Donor	`SCI-RTC-REQ-016`	When REQ-014 is not available, replacement shall fail under that route unless a separate donor-to-target authority declares direction, quantity/unit domain, support, uncertainty, validity, and provenance.
Donor	`SCI-RTC-REQ-017`	The despike detector, target indicator, candidate eligibility, topology, score, tie rule, donor weights, time relation, source-mask/flag use, fallback, and every realized donor link shall be reconstructible.
Donor	`SCI-RTC-REQ-018`	A replaced target shall retain target identity, link every donor and factor, preserve the replacement cause, and expose correlations from reused or shared donors; finiteness shall not create independent exposure.
Influence	`SCI-RTC-REQ-019`	Every output transitively influenced by an ALIGN-synthesized cell shall be scientifically ineligible while preserving its numerical effect, typed cause, and support.
Influence	`SCI-RTC-REQ-020`	Every output transitively influenced by an RTC-replaced cell shall be scientifically ineligible while preserving its numerical effect, typed cause, donor linkage, and support.
Filter	`SCI-RTC-REQ-021`	Every enabled response-changing stage shall record chronological order, exact full-precision coefficients or exact digest, coefficient convention, normalization, numeric precision, rate, direction, state, context, mask, edge, non-finite policy, and exceptions once per coherent segment.
Filter	`SCI-RTC-REQ-022`	An FIR claim shall expose exact impulse support, DC gain, phase convention, and normalization; a label such as “normalized” shall not substitute for the coefficient sum or response.
Filter	`SCI-RTC-REQ-023`	An IIR or notch claim shall expose an exact recurrence or state-space equivalent, realized poles/zeros where applicable, initial/final state, reset/carry boundary, transient/guard rule, precision, and stability status.
Mask	`SCI-RTC-REQ-024`	A processing mask shall be represented as an operator control, including geometry or selection, boundary convention, dilation, denominator/segment behavior, and state interaction; it shall not be treated as acquisition validity or confidence.
Mask	`SCI-RTC-REQ-025`	Invalid, unavailable, wrong-frame, out-of-domain, or ambiguously bound coordinates shall not become outside-source samples and shall fail the affected required coordinate operation.
Finite	`SCI-RTC-REQ-026`	A non-finite required sample, factor, coefficient, coordinate, or filter state shall be rejected, marked unavailable over its exact transitive influence, or handled by an explicitly authorized prior replacement/recovery rule; it shall not be silently coerced to zero.
Edge	`SCI-RTC-REQ-027`	Boundary extension, incomplete context, filter guards, IIR recovery, short/empty scans, and output selection shall be explicit; an edge or incomplete-state cell shall not be labeled an ordinary interior sample.

Area	ID	Normative statement
Sampling	SCI-RTC-REQ-028	V0.1 downsampling shall use only phase-zero point selection $y[n] = u[Mn]$; arithmetic-mean and other block-aggregate downsampling are excluded.
Sampling	SCI-RTC-REQ-029	Each coherent segment shall record M, zero phase, input/output rates and grids, the cardinality of equation (SCI-RTC-EQ-011), selected-point representative time, full transitive support, edge disposition, and flag/influence aggregation.
Sampling	SCI-RTC-REQ-030	Any alias-free or bounded-alias claim shall include the exact realized prefilter and all folded bands of equation (SCI-RTC-EQ-014), with a declared input band, metric, bound, and unavailable state; point selection alone shall not be called anti-alias filtering.
Sampling	SCI-RTC-REQ-031	Fixed mode shall resolve an admitted factor/filter/rate/compatibility plan before application and record it independently of realized output.
Learned	SCI-RTC-REQ-032	Learned mode shall preserve distinct requested, bootstrap, learned, resolved, and applied states and shall not advertise a learned plan before its admitted evidence and numeric policy are complete.
Learned	SCI-RTC-REQ-033	Learned resolution shall choose the largest admitted integer factor only after astronomical-transfer, exact phase-zero alias, sampling, filter-realizability, and downstream-compatibility constraints all pass using the smallest admitted beam and maximum valid in-scan speed.
Learned	SCI-RTC-REQ-034	The first learned applied scope shall use one common immutable observation cadence and filter across arrays and scans; percentile speed shall remain diagnostic rather than safety authority.
Learned	SCI-RTC-REQ-035	Apply shall consume exactly one immutable resolved plan and shall not retune, adapt, or silently fall back; an unavailable or rejected plan shall follow the selected failure/fallback policy before apply begins.
Learned	SCI-RTC-REQ-036	Restart shall restore a state-complete resolved plan, verify exact plan/input compatibility, and reject stale or incompatible beam, speed, scan, coefficient, rate, phase, role, or parent identity.
Response	SCI-RTC-REQ-037	The atomic bundle shall contain the complete realized response of SCI-RTC-DEF-010 or a typed unavailable state; a partial filter kernel shall not be labeled the conditioned response.
Response	SCI-RTC-REQ-038	Selection, detector mixing, time-varying calibration, masking, finite boundaries, state, or multirate behavior shall use a local/factorized detector-time response or truthful unavailability rather than an unjustified scalar transfer.
Response	SCI-RTC-REQ-039	A scalar LTI response shall be claimed only on a stated uniform, fixed-state, detector-separable, translation-invariant interior domain with no data-derived selector or undeclared rate change.
Response	SCI-RTC-REQ-040	Where defined, response shall distinguish amplitude, power, phase, group delay, sampling rate, Fourier convention, alias images, support, and response uncertainty; undefined phase/delay at a response zero shall remain unavailable.
Response	SCI-RTC-REQ-041	Output support shall be expanded transitively through alignment, replacement, every filter, state boundary, edge rule, and phase-zero selection; a selected center index shall not stand in for full support.

Area	ID	Normative statement
Statistics	SCI-RTC-REQ-042	Conditional mean and covariance shall follow equations (SCI-RTC-EQ-016a)–(SCI-RTC-EQ-016b) only for an admitted input model and fixed realized state, with reused donors and overlapping support represented as correlations.
Statistics	SCI-RTC-REQ-043	When selection depends on modeled random data, selection bias and selection covariance shall be modeled or explicitly unavailable; they shall not be set to zero merely by conditioning on the realized selector.
Statistics	SCI-RTC-REQ-044	Calibration, atmosphere, beam, donor-scale, coefficient, cadence/phase, response, and model nuisance terms shall retain value, validity, uncertainty, provenance, and correlation scope, including required cross terms.
Statistics	SCI-RTC-REQ-045	Unknown covariance shall remain unavailable and shall not be replaced by diagonal, white, stationary, independent, zero, or scalar-weight assumptions; conditional inverse variance shall not be called full precision or total uncertainty.
Validity	SCI-RTC-REQ-046	Acquisition support, original validity, ALIGN synthesis, numerical computability, coordinate validity, flags/causes, operator masks, replacement, transitive influence, complete support, response status, consumer eligibility, uncertainty availability, and provenance shall remain separate states.
Validity	SCI-RTC-REQ-047	A flag shall describe a cause and shall not automatically become a mask, invalidity, weight, or universal eligibility decision; aggregation shall preserve every contributing cause required by the selected policy.
Output	SCI-RTC-REQ-048	RTC shall produce the role-specific atomic bundle of equation (SCI-RTC-EQ-022); conditioned TOD alone shall not be reported as a complete required RTC product.
Output	SCI-RTC-REQ-049	Required malformed identity/state or missing authority shall propagate to atomic required-output failure; optional unavailable provenance detail shall be typed and numerically inert.
Output	SCI-RTC-REQ-050	Compact provenance shall reconstruct input/parent identity, detector binding, grid and intervals, calibration state, donor links, ordered filters, masks, coefficients/state/edges, sampling, response/uncertainty status, exceptions, counts, and all one-way state records.
Output	SCI-RTC-REQ-051	Each diagnostic shall name estimator, unit, support, mask/selection, validity, availability, and whether it is inert, advisory, or enters the selected policy; optional diagnostic detail shall not change values, flags, eligibility, cardinality, or failure.
Consumer	SCI-RTC-REQ-052	RTC shall supply cause-preserving validity inputs rather than a universal downstream eligibility bit; each consumer shall bind the exact RTC parent and shall not strengthen unavailable timing, coordinate, calibration, response, covariance, or eligibility claims.
Consumer	SCI-RTC-REQ-053	PTC-disabled terminal RTC output shall be a valid requested state when selected; no PTC or map product shall be implied, while any requested downstream product shall require the exact admitted RTC bundle.
Claims	SCI-RTC-REQ-054	Algebraic contract correctness, implementation conformity, representation fidelity, observational performance, science-impact qualification, and production readiness shall be stated and assessed as separate claims.

F Falsifiable predictions and limiting cases

Each prediction is an algebraic or semantic falsification target. A future conformance activity may choose an exact numerical tolerance justified by the operation and representation before observing candidate output. No result is reported by this document.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-001	Identity chain: raw role, $A = I$, no replacement or filter, $M = 1$, valid original samples. Values, unit, order, identities, cardinality, response, flags, and support are unchanged.	Any hidden calibration, value/order/cardinality change, flag loss, nonidentity kernel, or new claim of timing/coordinate authority.
SCI-RTC-PRED-002	Role separation: feed the same dimensionless raw Beammap input to raw and explicitly CAL-authorized requests. The raw output remains $\Delta f/f$; the calibrated output exists only with its admitted CAL state.	A raw product labeled mJy/beam, a calibrated product produced without CAL authority, or silent interchange between the products.
SCI-RTC-PRED-003	Raw donor direction: choose $\phi_q = 2$, $\phi_d = 4$, and donor $x_q = 3$. Equation (SCI-RTC-EQ-002) predicts raw target replacement 1.5 and equal calibrated amplitudes 6.	Use of ϕ_d/ϕ_q, output other than 1.5, unequal calibrated amplitude, or dependence on legacy `responsivity`.
SCI-RTC-PRED-004	Raw donor availability: independently make either factor invalid, convention/domain-incompatible, occurrence-mismatched, or set $\phi_d = 0$. The `flxscale` route is unavailable.	Finite replacement, stale factor reuse, row-number-only binding, zero insertion, or success without separate donor authority.
SCI-RTC-PRED-005	Order: compare calibration-before-replacement with a proposed reversed affine path using unequal factors and nonzero offsets. Outputs and local response agree only if equation (SCI-RTC-EQ-004) holds completely.	Equality asserted from multiplicative factors alone, ignored offsets/uncertainty, or fixture equality labeled a general proof.
SCI-RTC-PRED-006	Interior target/donor impulses under fixed selection. Each output column equals equation (SCI-RTC-EQ-012); donor energy appears in the declared detector column, support, scale, and phase.	Missing or circularly wrapped energy, unexplained mixing, wrong donor identity/direction, support, normalization, or phase.
SCI-RTC-PRED-007	Constant input with sufficient context. Output equals the constant times realized DC gain after the declared transient; a DC-preserving chain is exactly constant.	Drift, divide-by-zero mask normalization, edge values advertised as interior, or a false DC-preservation claim.
SCI-RTC-PRED-008	Interior step with complete context. Response equals the cumulative impulse response and settles according to DC gain and stability.	Undeclared overshoot, failure to settle, hidden state leakage, or disagreement with the impulse-derived response.
SCI-RTC-PRED-009	Affine ramp through each FIR and admitted alignment mapping. Output slope and offset follow the zeroth and first moments of the exact kernel.	Unexplained clipping, wrong phase/slope/offset, or a ramp-preservation claim without the required moments.
SCI-RTC-PRED-010	Long bin-centered and off-bin sinusoids at DC, passband interior/edge, transition, stopband, and near output Nyquist. Amplitude and phase follow equations (SCI-RTC-EQ-013) and (SCI-RTC-EQ-014).	Gain/phase outside preregistered tolerance, unrecorded frequency convention, or an aliased component labeled passband signal.
SCI-RTC-PRED-011	Sinusoids at the realized notch center and symmetric offsets under the exact rate and state. Depth, width, phase, and transient follow realized zeros/poles.	Center/depth/width mismatch, unstable pole, rate mismatch, hidden state, or an edge transient reported ordinary-valid.
SCI-RTC-PRED-012	Coordinate mask with valid inside/outside values plus invalid, unavailable, wrong-frame, out-of-domain, and ambiguously bound coordinates. Invalid classes fail the affected operation and remain distinct from outside.	Invalid becoming outside/unmasked, entering geometry, clamping to a center/edge, or loss of the distinct cause.
SCI-RTC-PRED-013	Samples just inside, exactly on, and just outside each selected mask boundary and temporal dilation. Membership follows the recorded inclusive/exclusive and dilation policy identically across execution modes.	An undocumented boundary, frame mismatch, asymmetric dilation, or sequence-dependent membership.

ID	Construction and prediction	Falsifier
SCI-RTC-PRED-014	Impulses and constants at first/last support cells and immediately inside complete FIR context. Values, guard, support, and output admission follow the selected edge rule.	Undeclared padding, truncation, wrap, renormalization, guard, or output selection.
SCI-RTC-PRED-015	Zero, impulse, and constant inputs split and rejoined at scan/chunk/state boundaries for an IIR. Results follow the declared reset/carry and terminal-state policy.	Prior-observation leakage, hidden carry/reset, unbounded non-finite poisoning, or contradiction between split and declared state semantics.
SCI-RTC-PRED-016	Lengths $0, 1, L-1, L, L+1$ around every support/guard threshold, with stable scan identity.	Crash, underflow, stale output, silent scan deletion/renumbering, or ordinary-valid status without required context.
SCI-RTC-PRED-017	Inject NaN, $+\infty$, and $-\infty$ separately into a sample, factor, coordinate, coefficient, and IIR state. Failure or unavailable influence follows REQ-026 exactly.	Silent zero/coercion, finite-looking output without cause, stale state recovery, or an unbounded tail reported valid.
SCI-RTC-PRED-018	Place one typed cause at every member of a phase-zero output's full support, including a non-center synthesized or replaced cell. Each cause propagates and either such influence makes the output scientifically ineligible.	Center-only aggregation, erased cause, finite output called eligible, or replacement/synthesis treated as independent exposure.
SCI-RTC-PRED-019	For $M = 1, 2, 3, 4, 5$ and $N = 0$ through several incomplete final strides, enumerate $j = Mn$. Cardinality and selected time follow equation (SCI-RTC-EQ-011); no block mean appears.	Wrong N_y, dropped/duplicated point, nonzero phase, averaged value, timestamp misidentification, or rate mismatch.
SCI-RTC-PRED-020	Two sinusoids separated by an integer multiple of f_{out}, before and after the realized prefilter. Folded amplitude is the sum in equation (SCI-RTC-EQ-014).	A folded term missing from response, inadequate stopband labeled alias-free, or incorrect image frequency/amplitude.
SCI-RTC-PRED-021	Reuse one donor for two targets and give neighboring outputs overlapping filter support. Equation (SCI-RTC-EQ-016b) predicts cross-target and temporal covariance when the admitted input covariance permits it.	Independent weights assigned, off-diagonal terms silently dropped, or reused donor described as additional exposure.
SCI-RTC-PRED-022	Enable a response-changing mask, donor mix, stateful filter, edge, and decimator while providing only a scalar FIR transfer. Complete response state is unavailable.	The partial scalar transfer labeled complete, or downstream use that assumes omitted stages are identity.
SCI-RTC-PRED-023	Learned candidates include at least two safe factors and one failing each astronomical-transfer, alias, sampling, realizability, and compatibility constraint. Equation (SCI-RTC-EQ-021) chooses the largest passing factor.	A smaller safe factor selected without rule, any failing factor applied, a percentile speed used for safety, or a missing numeric predicate treated as pass.
SCI-RTC-PRED-024	Serialize and restart a learned plan, then alter one of beam identity, speed support, scan set, coefficient precision, rate, phase, role, or parent. Apply rejects the mismatch.	Silent relearning, retuning, native fallback, stale-plan application, or realized state rewriting the request.
SCI-RTC-PRED-025	Execute two observations sequentially with distinct rates, constants, factors, masks, donors, states, and output parents.	Any coefficient, state, factor, donor, mask, rate, flag, or product link leaks from the first observation.
SCI-RTC-PRED-026	Toggle optional detailed provenance and inert diagnostics while holding the selected policy and realized operator fixed. Values, flags, influence, eligibility, cardinality, response, and failures are identical.	Any scientific or numerical result changes, or an observational/production claim appears from the toggle.

F.1 Analytic limits

The following limits are required: a compatible donor pair with $\phi_q = \phi_d$ gives unit raw donor scale; a replacement indicator tending to zero gives identity; filter coefficients tending to a Kronecker delta give identity; $M \to 1$ gives no rate change; a source-mask radius tending to zero selects only exact valid coincidences under the declared boundary; complete context approaches the declared interior response; and a causal pole approaching the unit circle exposes increasing memory and never retains a stability claim at or outside the circle.

End of shared normative core: `SCI-RTC-v0.1/r0.1`. View-specific material cannot add to or modify this scientific authority.